

Indian Institute of Technology Jodhpur
Department of Mathematics
Mathematics II MAL1020
Practice Set 2

AY 25-26

Semester II

1. Let $T : P_3 \rightarrow P_3$ be defined by

$$T(p(x)) = p(x) - p(1) - (x-1)p'(1).$$

With respect to the basis $B = \{1, x, x^2, x^3\}$:

- (a) Find the matrix representation $[T]_B$.
 - (b) Find a basis for $\ker(T)$.
 - (c) Find a basis for $\text{Range}(T)$.
 - (d) Compute the rank and nullity.
 - (e) Verify the Rank–Nullity Theorem.
 - (f) Show that $P_3 = \ker(T) \oplus \text{Range}(T)$.
2. Let $T : P_3 \rightarrow P_2$ be defined by

$$T(p(x)) = p'(x) - p'(0).$$

Consider the bases $B = \{1, x, x^2, x^3\}$ and $C = \{1, x, x^2\}$.

- (a) Find the matrix representation $[T]_{C \leftarrow B}$.
- (b) Determine $\ker(T)$.
- (c) Find $\text{Range}(T)$.
- (d) Compute rank and nullity.
- (e) Determine whether $\ker(T)$ is complementary to the subspace

$$W = \{p(x) \in P_3 : p(0) = 0\}.$$

3. Let $V = M_{2 \times 2}(\mathbb{R})$ and define

$$T(A) = A + A^T.$$

Consider the standard basis

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

- (a) Find the matrix representation $[T]_B$.
- (b) Find $\ker(T)$.
- (c) Find $\text{Range}(T)$.
- (d) Compute rank and nullity.
- (e) Show that the subspaces of skew-symmetric and symmetric matrices form complementary subspaces of V .

4. Let $T : P_2 \rightarrow P_2$ be defined by

$$T(p(x)) = xp'(x) + p(0).$$

Let $B = \{1, x, x^2\}$.

- (a) Find the matrix representation $[T]_B$.
 - (b) Find the characteristic of $\ker(T)$ and determine its basis.
 - (c) Find $\text{Range}(T)$.
 - (d) Compute the rank and nullity.
 - (e) Determine whether $\ker(T)$ and $\text{Range}(T)$ are complementary in P_2 .
5. Let $V = M_{2 \times 2}(\mathbb{R})$ and define

$$T(A) = A - \frac{1}{2} \text{tr}(A)I,$$

where I is the identity matrix.

- (a) Find the matrix representation of T with respect to the standard basis.
 - (b) Find $\ker(T)$.
 - (c) Find $\text{Range}(T)$.
 - (d) Compute rank and nullity.
 - (e) Show that $V = \ker(T) \oplus \text{Range}(T)$.
6. Let $T : P_3 \rightarrow \mathbb{R}^2$ be defined by

$$T(p(x)) = (p(0), p(1)).$$

- (a) Find the matrix representation of T with respect to the basis $\{1, x, x^2, x^3\}$.
- (b) Find $\ker(T)$.
- (c) Compute rank and nullity.
- (d) Find a subspace $W \subset P_3$ such that $P_3 = \ker(T) \oplus W$.

7. Let $T : P_3 \rightarrow P_3$ be defined by

$$T(p(x)) = p''(x).$$

- (a) Find the matrix representation of T in the basis $\{1, x, x^2, x^3\}$.
 - (b) Find $\ker(T)$ and $\text{Range}(T)$.
 - (c) Compute rank and nullity.
 - (d) Show that the subspace of polynomials of degree ≤ 1 is complementary to $\text{Range}(T)$.
8. Let $V = P_2$ and define two linear transformations

$$T(p(x)) = p(x) - p(0), \quad S(p(x)) = p(0).$$

- (a) Find the matrix representations of T and S .
- (b) Show that $T + S = I$.
- (c) Find $\ker(T)$ and $\ker(S)$.
- (d) Compute rank and nullity of both transformations.
- (e) Show that $P_2 = \ker(T) \oplus \ker(S)$.